
StateCut: Residual Attention Predicates and Exact Persistent Writes

Samuel Mausberg¹

Abstract

Can a deterministic decoder certify its next transition without rereading every historical key and value? We study sufficient conditions for preserving both the emitted token and every persistent write under an explicit numerical reference. The verifier evaluates signed attention residuals at rounding-cell boundaries, using mergeable count, sum, and range summaries; a sharp refinement retains the integer number of rows in the classical convex moment bound. A persistent forest supports adaptive refinement, while a transactional interval interpreter can certify persistent writes even when transient activations remain uncertain. We prove the residual, rounding, partition, and future-trace implications, and give a nonconstant family with zero raw-row reads after preprocessing. The finite-count refinement certifies a case that the chord rejects. On two pretrained checkpoints, sampled filtered heads match dense E24 through fallback, while no captured attention call is certified in full. Measured E24/backend differences and explicit formal-proof boundaries leave deployed backend equivalence and end-to-end acceleration open.

1. Introduction

During autoregressive decoding, historical key/value (KV) data are repeatedly read to compute attention. Query-dependent sparse methods reduce those reads by selecting a subset of the cache, with empirical quality evaluations (Tang et al., 2024; Ribar et al., 2024). We investigate a stricter contract: skip historical reads only when a verifier establishes the exact reference transition. Matching the current greedy token is insufficient because a small change to a new KV entry can affect later tokens. Consequently, the certificate must cover the persistent state as well as the immediate decision.

¹Research draft, not a conference submission. ¹StateCut Project. Correspondence to: Samuel Mausberg.

StateCut combines three ideas. First, it tests the residual $N - tD$ directly at an output rounding boundary t , preserving dependence between an attention numerator N and denominator D . Second, an append-only forest makes root summaries available without a compulsory scan of every fixed-size block. Third, it accepts a speculative computation when every live persistent write and the terminal greedy decision are exact. A failed attempt executes the reference from the original state.

The mathematical ingredients have substantial antecedents. Adaptive exact predicates are classical (Shewchuk, 1997); the first-moment chord bound is the Edmundson–Madansky inequality (Edmundson, 1957; Madansky, 1959); and neural-network interval propagation is an established verification method (Singh et al., 2019). We specialize and compose these ingredients for attention and persistent decoder state. We also derive and implement an exact finite-cardinality version of the moment envelope. This improves the previous continuous-mass relaxation without adding summary fields. We make no priority claim for the underlying convex extremal argument.

The executable target is a rational arithmetic profile with correctly rounded BF16 materializations. This choice makes algebraic reassociation legitimate and gives a small reference implementation against which to test certificates. It does not identify that profile with PyTorch, FlashAttention, or a pretrained checkpoint’s instruction-level behavior. Our contribution is a sufficient certificate and a falsifiable research artifact, with the remaining numerical and systems obligations stated explicitly.

2. Contract and Numerical Reference

Fix model parameters, operator semantics, legal inputs, visibility rules, and the tie rule. Let

$$F_R(C, x) = (y, C') \quad (1)$$

process a pending token x , return the predicted token y , and produce the state C' . The state contains every object that can influence a future invocation. In the prototype it includes the model identity, per-layer exact KV, token prefix, and position. A serving implementation additionally needs the relevant masks, position conventions, branch identity, slot

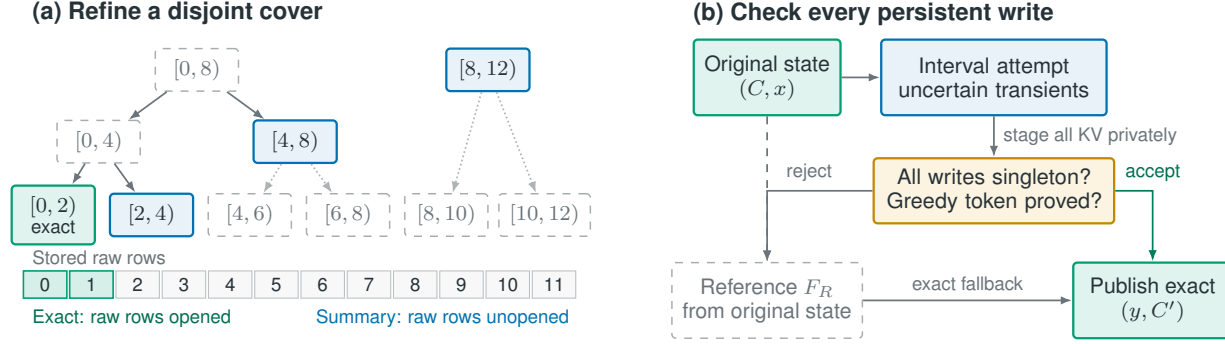


Figure 1. StateCut’s two boundaries. Left: a query refines only part of a persistent forest. Solid nodes form the current disjoint frontier; opening $[0, 2)$ reads two raw rows, while the other solid nodes retain summary enclosures. Dashed nodes are outside the current frontier. Right: uncertain transients remain private until every persistent write and the terminal decision are proved exact. Rejection invokes the reference on the original state. Colors are redundant with node labels and line styles.

ownership, generations, synchronization, and any recurrent or mutable state. These fields are part of the contract when they are live; a KV-only theorem cannot justify omitting them.

Why token equality is weaker. Consider $F(s) = (1[s < 0], s - 1/2)$. States $s = 1/4$ and $s = 3/4$ return the same first token. Their successors have different signs, so the next tokens differ. In contrast, equality of *both* outputs of Equation (1) composes by induction. Sampling would require its own coupling argument and an inventory of random state; the present policy is greedy with the smallest vocabulary index winning a tie.

Arithmetic profile. For rational scores, define

$$E_{24}(z) = 2^{-24} \text{RNE}_{\mathbb{Z}}(2^{24}e^z), \quad (2)$$

$$w_i = E_{24}(q^\top k_i), \quad A_j = \frac{\sum_i w_i v_{ij}}{\sum_i w_i}.$$

The query includes the exact attention scale. Dot products and reductions are rational; the materialization is $\text{BF16}(A_j)$ with nearest-even rounding. The profile canonicalizes signed zero and rejects nonfinite results. The exponential oracle refines rigorous Taylor enclosures until integer rounding is decided, with explicit resource limits and rejection on exhaustion. Weights can round to zero, so denominator positivity is a separate obligation. Matrix products sum exactly before each prescribed rounding cut. RMS normalization uses $\epsilon = 1/256$, and the gate is the specified $\text{BF16}(x/(1 + E_{24}(-x)))$. The full graph is defined by `src/statecut/model.py` and `docs/REFERENCE.md`.

We call this profile `RATIONAL_BF16_V1`. It is a deliberately explicit numerical object, not a claim of bitwise agreement with a common backend. In particular, E_{24} is not invariant under a common score shift: quantizing exponentials changes their ratios. PyTorch documents that

algebraically equivalent reductions can produce different floating-point results (PyTorch contributors, 2026). A mathematical softmax proof therefore needs a separate bridge before it can replace a pinned GPU instruction graph.

3. Residual Certificates from Mergeable Moments

A summary of $n > 0$ rows stores coordinatewise key minima and maxima and, for each value coordinate,

$$S = \sum_{i=1}^n v_i, \quad \ell \leq v_i \leq u. \quad (3)$$

The builder uses actual extrema; the theorem only requires valid bounds. Counts and sums add under merging, and extrema take minima and maxima. The reference also retains positive and negative moments for an independent interval intersection. No query-time raw scan is needed to merge summaries. Summaries must belong to the actual visible rows and the correct layer/head. Trusted construction is a premise, not a consequence of shape validation.

Interval dot products give $q^\top k_i \in [L, U]$ throughout a node. Since E_{24} is monotone, $a = E_{24}(L) \leq w_i \leq E_{24}(U) = b$. The same reasoning allows interval queries containing their reference values. Unsupported masks, biases, or visibility rules require additional sound score bounds; they are not implicitly covered.

For one value coordinate and threshold t , write

$$D = \sum_i w_i, \quad N = \sum_i w_i v_i, \quad (4)$$

$$R(t) = N - tD = \sum_i w_i (v_i - t),$$

and set $m = (a + b)/2$, $h = (b - a)/2$.

Lemma 1 (Residual from an absolute-sum envelope). *If $a \leq w_i \leq b$ and $\sum_i |v_i - t| \leq C(t)$, then*

$$m(S - nt) - hC(t) \leq R(t) \leq m(S - nt) + hC(t). \quad (5)$$

Proof. Subtract $m(S - nt)$ from Equation (4). The result is $\sum_i (w_i - m)(v_i - t)$, whose absolute value is at most $h \sum_i |v_i - t|$ by $|w_i - m| \leq h$ and the triangle inequality. $\square$

3.1. The Classical Chord Envelope

Convexity of $v \mapsto |v - t|$ gives, for $\ell < u$,

$$C_{\text{ch}}(t) = \frac{(nu - S)|\ell - t| + (S - n\ell)|u - t|}{u - \ell}. \quad (6)$$

For $\ell = u$, put $C_{\text{ch}}(t) = n|\ell - t|$. Then $\sum_i |v_i - t| \leq C_{\text{ch}}(t)$. This is the univariate Edmondson–Madansky first-moment bound applied to the empirical distribution (Edmondson, 1957; Madansky, 1959); Appendix A.1 gives the proof. It is sharp when arbitrary real mass is allowed at the two endpoints, although those masses need not correspond to integer row counts.

Both $S - nt$ and Equation (6) are unchanged by $(v_i, t) \mapsto (v_i + c, t + c)$. Thus a large common value offset does not itself enlarge the residual uncertainty. An independent quotient enclosure usually loses that cancellation.

3.2. A Sharp Envelope for an Integer Number of Rows

The continuous-mass relaxation discards information already present in the integer count n . Retaining it gives a stronger bound at constant summary cost. For $d = u - \ell > 0$, define

$$r = \frac{S - n\ell}{d}, \quad k = \lfloor r \rfloor, \quad x_* = \ell + (r - k)d. \quad (7)$$

Here $0 \leq r \leq n$. If $k < n$, put

$$C_{\text{int}}(t) = k|u - t| + (n - k - 1)|\ell - t| + |x_* - t|. \quad (8)$$

If $k = n$, use $C_{\text{int}}(t) = n|u - t|$; the case $d = 0$ uses $n|\ell - t|$.

Theorem 2 (Sharp finite-cardinality envelope). *For integer $n \geq 1$ and feasible (S, ℓ, u) ,*

$$\max_{\substack{v \in [\ell, u]^n \\ \sum_i v_i = S}} \sum_i |v_i - t| = C_{\text{int}}(t) \leq C_{\text{ch}}(t). \quad (9)$$

Consequently, Equation (5) with C_{int} gives the exact minimum and maximum of $R(t)$ over this value set and all independent choices $w_i \in [a, b]$.

Proof. If two coordinates are interior, preserve their sum and vary them in opposite directions. Their contribution to

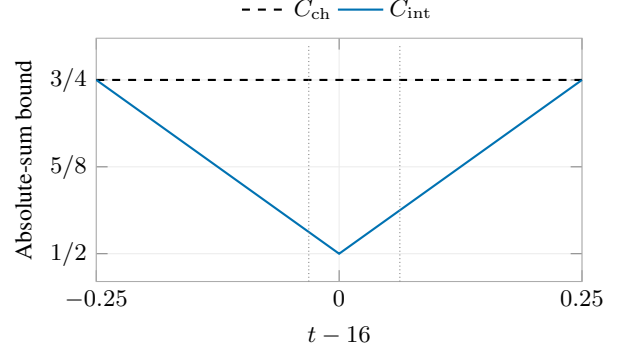


Figure 2. Exact analytic envelopes for three rows with sum 48 and range $[63/4, 65/4]$. The integer count forces the extremal vector $(63/4, 16, 65/4)$; a continuous-mass relaxation permits 1.5 rows at each endpoint. The vertical marks are the two BF16 boundaries of 16. This plot is a theorem illustration, not measured model data.

the objective is convex in that variation. One endpoint of its feasible interval has at least the original objective value and puts at least one coordinate at ℓ or u . Repeating leaves at most one interior coordinate. The fixed sum forces the endpoint counts and remaining value in Equations (7) and (8). This construction attains C_{int} , and the chord bound applied to it proves the inequality. Choosing $w_i = m \pm h \operatorname{sgn}(v_i - t)$ attains the two residual endpoints. Appendix A.2 details boundary cases and termination. $\square$

The sharpness is for the stated boxed relaxation. It does not retain key/value correlations, individual score constraints, discrete BF16 value feasibility, or discrete E_{24} weight feasibility. Those restrictions can make the true attention range smaller. For rational summary inputs, the extremal value vector is rational, so no irrational values are needed.

The refinement is translation invariant and never weaker than the chord bound. Its additive benefit is bounded:

$$0 \leq C_{\text{ch}}(t) - C_{\text{int}}(t) \leq (u - \ell)/2. \quad (10)$$

Thus the improvement per residual endpoint is at most $h(u - \ell)/2$; there is no implied factor improvement as n grows. For $n = 3$, $S = 48$, $\ell = 63/4$, and $u = 65/4$, the difference is strict at most interior thresholds (Figure 2).

3.3. Safe Intersections

Let $P = \sum_i \max(v_i, 0)$ and $M = \sum_i \min(v_i, 0)$. Then $N \in [aP + bM, bP + aM]$ and $D \in [na, nb]$. Interval subtraction of tD gives another residual enclosure. Intersecting it with Equation (5) is sound. A disjoint intersection signals a violated arithmetic or provenance premise and must reject. The direct threshold predicate itself requires no division by an uncertain denominator; computing the moment parameters still uses scalar divisions by known summary values.

4. From Residual Signs to Exact Decisions

For a proposed finite output y , let J_y be its exact rounding cell, with boundaries t_- and t_+ and separately specified endpoint inclusion. Ordinary BF16 nearest-even cells include both midpoints for even significands and exclude both for odd significands. The canonical zero cell is $[-2^{-134}, 2^{-134}]$; maximal finite values have an open overflow boundary. Neighbor spacing can be asymmetric at powers of two.

Theorem 3 (Residual rounding gate). *Suppose $D > 0$. Let $\underline{R}(t_-)$ and $\overline{R}(t_+)$ be sound lower and upper residual bounds. If*

$$\underline{R}(t_-) \geq 0, \quad \overline{R}(t_+) \leq 0, \quad (11)$$

with a strict comparison at each excluded endpoint, then $\text{BF16}(N/D) = y$.

Proof. The identity $R(t) = D(N/D - t)$ and $D > 0$ convert each residual sign to the corresponding cell-membership inequality. Membership in J_y is precisely the definition of rounding to y . $\square$

For a disjoint frontier of nodes, sum residual bounds at the *same* threshold. An opened leaf contributes its exact $N - tD$. The proposal is untrusted: a bad candidate may cause rejection, but cannot create a false certificate under the stated premises. The reference proposer uses midpoint weights and summary means; its work is charged to the filter.

A centered quotient $t + R(t)/D$ is sound when denominator division is valid, and is useful for propagating transient intervals. It can nevertheless remain too wide: its independently combined extrema need not be jointly attainable. The sign checks in Equation (11) retain the dependence needed for rounding without constructing that quotient.

Proposition 4 (Nonconstant family with no raw reads). *Let $n > 0$ be even, with half the values $63/4$ and half $65/4$. For any weights in $[7/8, 9/8]$, $\text{BF16}(N/D) = 16$, and the chord residual gate proves this from one summary.*

Proof. Here $S = 16n$, $m = 1$, and $h = 1/8$. The cell of 16 is the closed, asymmetric interval $[511/32, 257/16]$. Both boundaries are inside the value range and $C_{\text{ch}} = n/4$ there. The lower-bound residual is $n/32 - n/32 = 0$ at the left endpoint; the upper-bound residual is $-n/16 + n/32 = -n/32 < 0$ at the right. Finally $D \geq 7n/8 > 0$. $\square$

This is an existence result after preprocessing. The illustrative weight box is not the same as the $E_{24}(\pm 1/8)$ box in the executable balanced fixture. That fixture is separately checked against its actual quantized weights. No acceptance

Algorithm 1 Exact attention cut from a disjoint frontier

Require: Trusted forest, query q , expansion budget K

- 1: $\mathcal{F} \leftarrow$ roots and tail; all pieces unresolved
- 2: **for** $s = 0, \dots, K$ **do**
- 3: Bound each unresolved piece's weights; propose y
- 4: Sum denominator and residual bounds over $\mathcal{F}$
- 5: **if** denominator is positive and every coordinate passes Theorem 3 **then**
- 6: **return** y with an accepted-cut certificate
- 7: **end if**
- 8: **if** $s = K$ or no unresolved pieces remain **then**
- 9: **break**
- 10: **end if**
- 11: Choose an unresolved piece using enclosure width
- 12: **if** it is internal **then**
- 13: Replace it by its two children
- 14: **else**
- 15: Read its raw rows and store its exact contribution
- 16: **end if**
- 17: **end for**
- 18: Exactify only unresolved pieces; reuse exact contributions
- 19: **return** exactly rounded dense attention under `RATIONAL_BF16_V1`

rate for pretrained activations follows from either construction.

5. A Persistent Refinement Forest

Raw leaves contain at most B consecutive entries. Every full leaf is sealed and merged with adjacent equal-rank roots, as in a binary counter. A partial tail is separate. For $m = \lfloor N/B \rfloor$ full leaves there are $\text{popcount}(m)$ completed roots and at most one tail, hence $O(\log(2 + N/B))$ initial summaries. Raw entries are retained; the index does not establish memory compression. Figure 1 illustrates a partially refined frontier.

Theorem 5 (Partition and raw-read accounting). *The frontier in Algorithm 1 remains a disjoint cover of the visible prefix. Each raw leaf is read at most once, including exploration and fallback, under the exact-rational reference.*

Proof. Append and adjacent binary carries preserve consecutive disjoint coverage. A split replaces $[a, c)$ by $[a, b)$ and $[b, c)$ with the same union. Exactification changes only representation. Exact pieces are never opened again, and fallback descends only through unresolved pieces; those cannot overlap exact ones. Thus no raw leaf is read twice. $\square$

A root-only attempt requires $O((d_k + d_v) \log(2 + N/B))$ scalar summary work, and $O(d_k + d_v)$ for a single root.

This is a logical operation count; rational bit complexity, allocation, and interpreter overhead are outside that count. The current Python scheduler scans the frontier for priority on each expansion. The tree can still be fully explored, followed by all raw reads; a budget limits speculative exploration, not fallback work.

Across m completed leaves, fewer than m binary merges occur, so merges are amortized constant per completed leaf. Summary updates and initial construction must still be charged. Python root tuples and the toy token prefix are copied, with the latter costing $O(T)$ per append. A backend that fixes a nonassociative reduction tree may require a fresh dense pass after failed speculation; Theorem 5 does not remove that cost.

6. Certifying the Persistent Write Frontier

Some consumers erase transient uncertainty. Let $W(h)$ denote every new persistent write and $G(h)$ the terminal decision. It is sufficient to prove $W(\hat{h}) = W(h_R)$ and $G(\hat{h}) = G(h_R)$ even when $\hat{h} \neq h_R$. StateCut’s `writes` strategy realizes this observation using a sound interval interpreter, exact rounding, and private staging. It requires singleton rounded K/V coordinates before appending to a staged forest; queries and attention outputs may remain intervals. The `cuts` strategy instead certifies exact attention materializations and executes the exact suffix, using a decision gate only where no later persistent writes depend on uncertain transients.

Lemma 6 (Greedy decision with ties). *Suppose $z_j \in [L_j, U_j]$. Candidate c is the smallest-index maximizer if*

$$U_j < L_c \ (j < c), \quad U_j \leq L_c \ (j > c). \quad (12)$$

Proof. For an earlier competitor, $z_j \leq U_j < L_c \leq z_c$; for a later competitor, $z_j \leq U_j \leq L_c \leq z_c$. These are exactly the required strict and weak comparisons. $\square$

Theorem 7 (Exact transition and future traces). *Assume every represented interval contains its reference intermediate, every newly persistent scalar has a singleton interval, and the full vocabulary satisfies Equation (12). Commit all staged writes together. The resulting pair (y, C') equals Equation (1). If every rejected attempt runs F_R from the original state, the filtered decoder has the same tokens and persistent state for every finite legal continuation.*

Proof. A reference write contained in $[a, a]$ equals a by antisymmetry. Coordinatewise equality covers every new persistent write; unchanged committed fields remain equal. Lemma 6 proves token equality. An accepted step and a rejected step therefore both equal F_R . Induct over supplied input tokens. For autonomous decoding, use state (C, x) and feed the common output y as the next pending input. $\square$

Table 1. Balanced fixture: raw rows / summary records read per query. All three methods agree with dense E24. Initial indexing is additional work.

Rows	Indep. flat	Centered ratio	Direct
128	128/4	128/1	0/1
512	512/16	512/1	0/1
2,048	2048/64	2048/1	0/1
8,192	8192/256	8192/1	0/1

The interpreter uses signed interval affine operations, sound interval squaring and square roots, and monotone BF16 rounding. It multiplies an interval for x by an interval for $1/(1 + E_{24}(-x))$; it does not assume the entire SiLU-like function is monotone. Positive RMS epsilon prevents a zero normalizer. Appendix C explains the containment induction and its trusted components. Any ambiguous write or decision, invalid denominator, or resource failure discards the attempted transition before dense fallback. There is no partial commit or future repair debt.

7. Evaluation and Evidence Scope

We test three distinct questions: whether a predicate accepts fewer raw reads while preserving the named reference, whether complete toy-decoder states remain equal, and whether device or pretrained observations expose obstacles to deployment. The first two use exact-rational oracles. Device tests and pretrained captures are separately scoped in the generated measurement record below. All tabulated logical-read data are read from machine-readable experiment artifacts by `paper/render_results.py`.

7.1. Paired Logical-Read Ablations

The balanced fixture uses repeated four-way combinations of keys $\pm 1/8$ and values $16 \pm 1/4$, query $q = 1$, and leaves of $B = 32$. It compares the original independent numerator/denominator gate, a centered-quotient tree gate, and the direct residual tree gate on identical rows with zero expansion budget. Every result is checked against dense RATIONAL_BF16_V1. Table 1 reports raw reads and summary reads at four lengths. Index construction reads the entire prefix and is excluded from query reads, not from total system cost.

The direct gate accepts the balanced cases from one summary. Both divided variants require dense fallback. This separates the threshold predicate from merely centering a quotient, and the tree from the original flat summary scan. A separately seeded unstructured case retains full raw fallback; there is no universal sublinear claim.

Finite-cardinality improvement. For $n = 3$, $S = 48$, $\ell = 63/4$, $u = 65/4$, and score bounds $[-3/20, 3/20]$, the

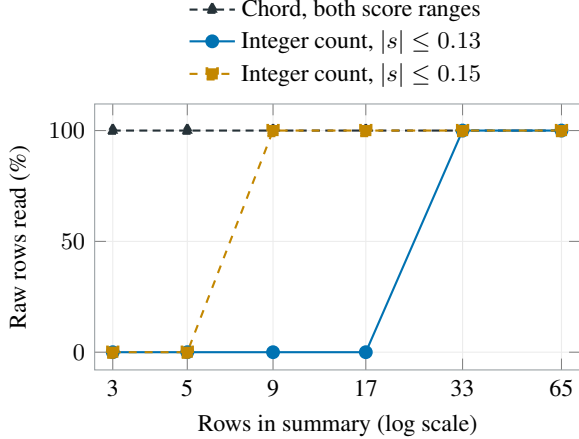


Figure 3. Measured raw reads in the finite-count ablation. Each point is one deterministic odd-length fixture with fixed score magnitude, zero expansion budget, and a single summary. The chord gate reads all rows. Finite-count gains disappear at larger counts, consistent with the additive bound in Equation (10). Lines connect tested points, not a fitted model.

integer-count residual gate can pass the cell of 16 where the chord residual gate fails. The experiment fixes the same summary, proposal, and actual E_{24} weight endpoints for both bounds. Sharpness tests enumerate small rational grids and endpoint weights, and include translation, degeneracy, and exact-integer input cases. The recorded lower-bound residuals at the left cell boundary are

$$\underline{R}_{\text{ch}} = -\frac{4862889}{268435456} < 0, \quad \underline{R}_{\text{int}} = \frac{497277}{33554432} > 0. \quad (13)$$

Both methods return the dense result 16; the chord method reads 3 raw rows and the integer method reads 0. An exhaustive finite-grid audit checks 5,004 row multisets, 35,028 row/threshold combinations, and 1,218 fixed-sum/threshold optima against both attained residual extremes. It uses counts 1–6 and the nine-point grid from -1 to 1 in increments of $1/4$; these finite tests support the implementation, while the general bound is proved mathematically.

Figure 3 retains both the favorable finite-count cases and the cases where both methods fall back.

7.2. Complete-State and Write-Frontier Checks

Four small random decoders (seeds 0, 1, 8, 19) each generate 24 transitions under both strategies, with expansion budget 3 and leaf size 4. The harness compares every returned token and complete state to dense execution. These trajectories are deterministic constructed tests, not independent samples from an LLM workload. The write-frontier acceptance rate is 7 of 96 steps; broad interval propagation often fails. A separate two-layer construction permits a non-singleton attention materialization yet certifies all per-

Table 2. Complete-state toy-decoder comparisons. Raw reads sum over layers and steps; write acceptance applies only to the writes strategy. Every compared token and full state agrees.

Strategy	Steps	Raw rows	Write accepts
Cuts	96	588	—
Writes	96	2,254	7

Table 3. Pretrained observer transparency, compared to the unchanged backend. State comparisons include prefill and every decode step; events are captured decode attention calls. This does not run a filtered replacement.

Model	Cases	Steps	States	Events
Qwen2.5-0.5B	6	18	24	432
SmolLM2-135M	6	18	24	540

sistent writes and the decision without historical raw reads. Its second-layer K/V projections are zero, so it establishes logical separation rather than typical model behavior.

7.3. Hardware, Pretrained Captures, and Formal Build

The GH200 device arithmetic oracle check passed 68,851 comparisons: 2,500 moment cases, 65,279 finite canonical BF16 cells, 1,025 E24 weight cases, 42 lattice-rounding cases, and 5 negative-tail cases. These are executed finite comparisons, not a formal proof of device arithmetic. The CUDA checker uses the chord envelope; the finite-count refinement is used by the rational CPU reference.

On NVIDIA GH200 480GB (compute capability 9.0), a synthetic 8-head, 32,768-row, 64-coordinate root gate has a $62.85 \mu\text{s}$ median single-call event span ($67.62 \mu\text{s}$ empirical 95th percentile), including host submission gaps. Graph batches of 100 sequential fixed-input calls give $45.84 \mu\text{s}$ median per call ($45.85 \mu\text{s}$ 95th percentile); the separate SDPA primitive gives $32.27 \mu\text{s}$ in that graph protocol. Recorded construction took 144.68 ms. The proposal was the construction’s known value 16, and the accepted gate read zero raw KV rows. Torch 2.7.0 and CUDA 12.8 were used. Graph amortization is not token latency. The two primitives have different numerical contracts; construction, refinement, and a decoder suffix are excluded from their event spans.

The observer study uses pinned checkpoint revisions, three deterministic prompt constructions at contexts 128 and 512, and three decode steps per case. All recorded per-layer KV tensors and final-position logits agree bitwise with the unobserved run. It tests capture transparency; it does not test StateCut as a model replacement. Checkpoint hashes and exact prompt token IDs are retained in the source records.

Pretrained local-gate acceptance. The CUDA audit evaluates 972 captured attention calls and 10,908 query heads per configuration across both checkpoints. It compares a

Table 4. Pretrained CUDA acceptance: 10,908 heads and 972 calls per configuration. Mean uses the lattice enclosure. Oracle columns compare the half-quantum (HQ) and lattice (L) weight enclosures. The last column counts complete calls for mean/oracle with L.

Block	Mean	Oracle HQ	Oracle L	Calls
1	0	254	280	0/0
16	0	0	0	0/0
64	0	0	0	0/0
128	0	0	0	0/0

summary-mean proposal with the already observed SDPA output as a diagnostic oracle proposal. The oracle’s dense computation is additional work, and its output need not equal E24.

Monotone rounding of outward exponential endpoints onto the E24 lattice tightens the half-quantum enclosure (Appendix D) and changes block-1 oracle acceptance from 254 to 280 heads. No complete all-head call passes. Block size 1 reads one summary per row, so its local certificates supply no sublinear summary-read advantage. Domain rejection and failure to prove positive mass are counted separately in Appendix G. These fixed workloads provide negative evidence for the current coarse summaries, not a population acceptance estimate.

Exact reference on captured heads. A deterministic subset evaluates 72 query heads (4,608 coordinates): the first two heads in the first, middle, and last layer at the first decode step for each model/prompt/context case. The finite-count CPU filter agrees with dense E24 on every tested head, using all raw rows through fallback in every case. E24 matches the observed SDPA bits on 4,107 coordinates and differs on 501. This comparison uses E24’s canonical zero and the unmodified observed bits. The measured disagreement rules out treating these numerical profiles as interchangeable. One Qwen head has actual positive scores from 1,237.073 to 1,280.256; extending the rigorous CPU exponential oracle to 2,048 with magnitude-aware precision supports the tested cases. No common score shift is applied.

Capture-only evaluation preserves the selected backend’s returned output and records actual query, key, value, and output tensors. Re-evaluating those tensors under RATIONAL_BF16.V1 asks whether the profiles happened to agree on the observed input. It cannot prove a universal numerical bridge, and observing one attention layer cannot establish a complete decoder transition. A CUDA gate microbenchmark measures the supplied local certificate; its comparison to a separate SDPA primitive does not measure an integrated decoder speedup.

The pinned Lean build succeeds and audits 66 source theorems (84 compiled theorem constants including generated

helpers). Only the standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound` occur. The compiled results cover chord and finite-count residual bounds, finite-count attainment and chord dominance, a mathematical floor/remainder contract, open/closed cell gates, disjoint-sum composition, singleton writes, and future traces. Appendix H maps the claims to actual compiled declaration names. The proof does not establish a Python/CUDA refinement or instantiate a deployed-backend error bound. The compiled declarations establish real-valued algebra and abstract state composition. The complete finite-format interpreter and generated GPU code remain outside that proof. The trusted base includes Python integers and fractions, interval and rounding code, summary construction and attribution, and compiler/device semantics for the CUDA experiment. A successful proof build does not verify these implementations by itself.

8. Related Work

Filtering and moments. Shewchuk (1997) established adaptive exact numerical predicates for geometric decisions. The general accept/refine/fallback pattern predates this work. The chord in Equation (6) is classical (Edmundson, 1957; Madansky, 1959); finite-vector convex extremality fits the majorization framework (Marshall et al., 2011). StateCut applies those tools to $N - tD$ at representable output boundaries and connects the result to persistent writes.

Attention selection and certification. FlashAttention reduces dense attention memory traffic through tiling (Dao et al., 2022). Quest already stores key extrema to select query-relevant pages (Tang et al., 2024); SparQ reduces cached-history transfers by selective fetching (Ribar et al., 2024). Bounds for selecting attention blocks also precede this manuscript (Butucea, 2026). A bound on a block’s best score alone does not certify its omission from a normalized output. We additionally certify the output cell using value moments and all unresolved mass.

Recent preprints give local quantized-attention error bounds and fallback (Calver, 2026), deterministic and probabilistic runtime witnesses (Wei et al., 2026b), and contracts for heterogeneous memory updates and reads (Wei et al., 2026a). These are related contributions with their own stated evidence boundaries; we do not claim invention of runtime attention certification or state-aware memory reasoning. Our contract requires exact newly persistent values under one numerical profile, and our main construction certifies read omission through residual signs.

Verification and generation. Abstract interpretation has long supported neural-network certificates (Singh et al., 2019). An existing public research artifact applies exact-

real interval certification to full-vocabulary next-token decisions for pretrained language models (Naughton, 2026). StateCut’s persistent write condition is needed to reuse a speculative transition without altering future state. Speculative decoding preserves a sampling distribution through proposal verification (Leviathan et al., 2023); that contract differs from deterministic token-and-state equality. Conditional lower bounds for general attention (Duman Keles et al., 2023) are consistent with a preprocessed sufficient condition that can fail on every difficult query. We establish no worst-case acceleration theorem.

9. Limitations and Required Backend Bridge

The remaining systems question is whether certificates are both useful and cheaper than the work they avoid. BF16 cells near zero are narrow; large key boxes, mixed values, and many coordinates can make full-cut acceptance rare. Layerwise interval propagation compounds this difficulty. The finite-count improvement has bounded additive gain. Raw KV storage, summary maintenance, complete vocabulary checks, and dense fallback remain necessary. This artifact does not establish competitive end-to-end performance, general pretrained acceptance, or a serving-ready integration.

For a pinned backend output B and mathematical attention A , a possible bridge proves $|B - A| \leq \varepsilon$ for the entire relevant instruction graph. Then certify A inside $[t_- + \varepsilon, t_+ - \varepsilon]$, with the appropriate strict edges. Appendix E proves the implication; obtaining ε must account for products, reductions, exponentials, normalization, fusion, conversion, and exceptional values. A measured maximum error is not a universal premise. NVIDIA explicitly identifies certain mathematical-function error tables as non-exhaustive observations (NVIDIA, 2026). The local CUDA checker uses directed arithmetic and its own exponential enclosure; it does not supply this backend bridge.

For dense cost T_D , filter/proposal cost T_F , maintenance cost T_M , accepted suffix cost T_A , and full valid acceptance probability p , a simple serial model gives

$$\mathbb{E}[T] = T_F + T_M + pT_A + (1 - p)T_D. \quad (14)$$

A speedup requires $T_F + T_M < p(T_D - T_A)$. This is a break-even condition, not an estimate of p . Initial indexing, transfers, page migration, launches, synchronization, retries, and unaffected model work must be charged in an integrated benchmark.

Impact Statement

This work studies the reliability of computation-saving inference methods. Exactness claims with incomplete numerical or state contracts can conceal behavior changes, so the

artifact makes those contracts and their remaining implementation gaps explicit. Lower inference cost, if ultimately achieved, would inherit the benefits and misuse risks of the models being run. The present work does not evaluate downstream safety, fairness, or societal outcomes and does not make a deployment recommendation.

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A. Complete Mathematical Proofs

All sums below are finite. Real-valued theorem statements include the rational reference as a special case. None assumes statistical independence of keys, values, or queries. The only independence used for sharpness is freedom to choose each weight anywhere in its stated box.

A.1. Convex Chord Bound and Continuous-Mass Sharpness

Suppose $\ell < u$ and $v \in [\ell, u]$. Put $\alpha = (u - v)/(u - \ell)$ and $\beta = (v - \ell)/(u - \ell)$. Then $\alpha, \beta \geq 0$, $\alpha + \beta = 1$, and $v - t = \alpha(\ell - t) + \beta(u - t)$. The triangle inequality gives

$$|v - t| \leq \alpha|\ell - t| + \beta|u - t|. \quad (15)$$

Summing and substituting $S = \sum_i v_i$ yields Equation (6). For $\ell = u$, all $v_i = \ell$, so the expression is exact. This proves soundness without relying on a probabilistic interpretation of the moment.

For nonnegative real masses with total n and first moment S , masses $\lambda_\ell = (nu - S)/(u - \ell)$ and $\lambda_u = (S - n\ell)/(u - \ell)$ at ℓ, u are feasible and attain C_{ch} . They are integers only for particular summaries. Choosing weights $m + h \operatorname{sgn}(v - t)$ or $m - h \operatorname{sgn}(v - t)$ attains the corresponding residual extreme for that endpoint distribution. This proves sharpness for the continuous-mass relaxation, not generally for n integer rows.

A.2. Finite-Cardinality Extremality and Its Quantitative Gain

We prove Theorem 2 constructively. The cases $\ell = u$, $S = n\ell$, and $S = nu$ force a constant vector and agree with the stated formula. Assume otherwise that $d = u - \ell > 0$. Start with an arbitrary feasible vector v . If two coordinates x, y lie strictly inside (ℓ, u) , define

$$I = [\max(\ell - x, y - u), \min(u - x, y - \ell)]. \quad (16)$$

This is the feasible interval for replacing (x, y) with $(x + \delta, y - \delta)$. Its lower endpoint is strictly negative and its upper endpoint strictly positive, so 0 lies inside I . The function $f(\delta) = |x + \delta - t| + |y - \delta - t|$ is convex. Writing 0 as a convex combination of the endpoints shows that at least one endpoint δ_* has $f(\delta_*) \geq f(0)$. At that endpoint at least one of the two coordinates equals ℓ or u . No previously fixed endpoint coordinate changes, the sum is preserved, and the objective does not decrease. Each operation reduces the number of interior coordinates by at least one, so at most $n - 1$ operations leave at most one interior coordinate.

If the resulting vector has one interior coordinate x , let k be the number of upper endpoints. Then

$$S = ku + (n - k - 1)\ell + x = n\ell + kd + (x - \ell). \quad (17)$$

Since $0 < x - \ell < d$, $k = \lfloor (S - n\ell)/d \rfloor$ and $x = x_*$. If no coordinate is interior, $r = (S - n\ell)/d$ is an integer. For $r < n$ the formula with $x_* = \ell$ counts that endpoint once; for $r = n$ the separate all- u case applies. In every case the final objective is exactly $C_{\text{int}}(t)$. Because it is no smaller than the arbitrary initial objective, it is an upper bound. The described final vector is itself feasible, so the bound is attained. This proves the maximum equality in Equation (9). Applying the chord inequality to the final vector proves $C_{\text{int}} \leq C_{\text{ch}}$.

For residual sharpness, Lemma 1 supplies the bound. On a maximizing value vector choose $w_i = m + h \operatorname{sgn}(v_i - t)$ to obtain

$$\sum_i w_i(v_i - t) = m(S - nt) + h \sum_i |v_i - t|. \quad (18)$$

These weights lie in $[a, b]$; at $v_i = t$ any allowed weight suffices. Changing the sign before h attains the lower endpoint. The denominator may vanish if the box includes zero, but the residual extremality theorem does not divide by it. Positivity is separately required for rounding.

To prove Equation (10), observe that the chord is exact at all endpoint coordinates of the extremal vector. Its excess therefore comes only from x_* . If $t \notin (\ell, u)$, absolute value is affine on $[\ell, u]$ and the excess is zero. Otherwise let $g(x)$ be the chord through $(\ell, |\ell - t|)$ and $(u, |u - t|)$. For $x \leq t$,

$$g(x) - |x - t| = \frac{2(x - \ell)(u - t)}{d}, \quad (19)$$

and for $x \geq t$ it equals $2(u - x)(t - \ell)/d$. The maximum occurs at $x = t$ and is $2(t - \ell)(u - t)/d \leq d/2$, because $(t - \ell)(u - t) \leq d^2/4$. Thus a single remainder coordinate accounts for all the improvement and establishes its additive upper bound.

A.3. Translation Invariance

Replace v_i, ℓ, u, t, S by $v_i + c, \ell + c, u + c, t + c, S + nc$. Then $u - \ell, S - n\ell, nu - S$, and $S - nt$ are unchanged. The translated r and k are unchanged, and x_* becomes $x_* + c$. Every absolute difference in both envelopes is unchanged. The exact residual is unchanged term by term. Therefore both the chord and finite-cardinality residual enclosures are translation invariant. This statement keeps weights fixed; it does not claim invariance under score shifts in Equation (2).

A.4. Denominator Positivity, Cell Edges, and Proposal Soundness

For each disjoint node b , a nonnegative weight box gives $D_b \in [n_b a_b, n_b b_b]$. Summing the lower endpoints and obtaining a strictly positive result establishes $D > 0$. An exact opened leaf may instead contribute a known positive denominator. A zero lower bound is inconclusive even when the actual denominator is positive; a verifier may refine, but cannot skip this premise.

For an arbitrary interval cell J_y with independently open or closed edges, the lower test is $\underline{R}(t_-) \geq 0$ if the edge is included and $\underline{R}(t_-) > 0$ otherwise. The upper test similarly uses $\overline{R}(t_+) \leq 0$ or < 0 . The identity $R(t) = D(A - t)$ converts these to the exact membership conditions when $D > 0$. Every possible A belongs to the same cell, so rounding is decided independently of how the proposal was obtained.

BF16 has subnormal spacing 2^{-133} , making the canonical zero cell $[-2^{-134}, 2^{-134}]$. A normal power of two has unequal spacing to its predecessor and successor, so replacing its true cell with a symmetric ULP window can be unsound. At 16, the predecessor is $255/16$ and the successor is $129/8$; their midpoints with 16 are $511/32$ and $257/16$. The largest finite positive magnitude has odd significand and an open overflow midpoint under this finite-only profile. Negative cells are reflected from positive cells. The executable test enumerates all finite canonical BF16 centers and checks these cell memberships directly against exact rounding.

A.5. Sound Summaries and Interval Queries

Let a query coordinate satisfy $q_j \in [\underline{q}_j, \overline{q}_j]$ and a stored key coordinate satisfy $k_{ij} \in [\underline{k}_j, \overline{k}_j]$ for every row. The product belongs to the interval with endpoints the minimum and maximum of the four endpoint products. Summing these intervals gives an enclosure $[L, U]$ of every dot product. Exact scaling and any sound bias enclosure may be incorporated before applying the nondecreasing E_{24} . Integer nearest-even rounding is nondecreasing, as is e^z , hence $E_{24}(L) \leq E_{24}(q^\top k_i) \leq E_{24}(U)$. If a loose summary score interval exceeds the oracle resource domain, the filter may open nodes or fall back to legal raw scores; it must not infer that the reference invocation itself is invalid merely from the loose box.

Summary merging follows directly from finite-sum associativity and extrema over disjoint unions. For the positive/negative intersection, each positive v_i contributes in $[av_i, bv_i]$ and each negative v_i in $[bv_i, av_i]$. Summing yields $[aP + bM, bP + aM]$. The denominator interval, signed interval scaling by t , and subtraction are sound operations. Intersecting two sets containing the same residual preserves containment. A disjoint result contradicts the premise that both sets contain it.

B. Forest Invariants and Complexity Details

Suppose the current completed roots cover consecutive disjoint ranges and have strictly descending ranks. Appending a new full leaf creates a rank-zero root at the right end. Merging the last two roots when they have the same rank preserves their union, contiguity, and all summaries. Repeating the carry produces exactly the set bits of the completed-leaf count. Induction from the empty prefix proves root coverage. The partial tail, if nonempty, starts at the end of the completed prefix and ends at N , completing the partition.

If there are m completed leaves and $p = \text{popcount}(m)$ roots, exactly $m - p$ merges have occurred: each completion increases the number of roots by one and each merge decreases it by one. In particular, $m - p < m$ for $m > 0$. One completion can still trigger $\lfloor \log_2 m \rfloor + 1$ carries. The summaries contain $O(d_k + d_v)$ scalars, and a full binary tree with m leaves contains fewer than $2m$ nodes. With the partial tail, metadata storage is $O((1 + N/B)(d_k + d_v))$ scalar slots, in

addition to $O(N(d_k + d_v))$ raw data. These counts suppress integer/rational bit size.

For a query, replacing a node by disjoint children preserves both coverage and sum of exact contributions. Each piece is marked unresolved or exact; only unresolved pieces can be selected for refinement. Because descendants of disjoint pieces are disjoint, fallback through unresolved subtrees never revisits an exactified raw leaf. This is the operational condition underlying Theorem 5, not merely a property of the tree shape.

C. Interval Interpreter and Transactional Composition

The interpreter claim is conditional on a specified computation graph and sound primitive implementations. For any acyclic reference graph, prove containment in topological order. Point inputs are enclosed exactly. An interval affine map uses signed endpoint products and interval addition; these contain the corresponding exact products and sums. Interval squaring uses zero as its lower endpoint exactly when zero lies in the input interval. For a nonnegative radicand interval, monotonicity of the square root allows outward enclosures of its two endpoints. Reciprocal reverses endpoints on a strictly positive interval. These facts preserve RMS normalization containment after adding $\epsilon > 0$.

The profile’s exact point RMS calculation decides BF16 rounding by rational square comparisons for $x/\sqrt{r}$; it need not approximate the irrational number first. For non-point arguments, a square-root interval and outward division suffice. The sigmoid factor $1/(1 + E_{24}(-x))$ is monotone, but $x/(1 + E_{24}(-x))$ need not be. Multiplying enclosures of the two factors therefore supplies the safe rule. Whenever a represented result is materialized, monotonicity of the finite BF16 rounding map sends a real interval to an interval of possible rounded values. If this interval is a singleton, it equals the reference materialization. Overflow or an unresolved numerical guard ends speculation.

Attention bounds contain all reference query/key/value combinations represented by the current intervals. Their division requires a positive denominator enclosure. Residual gates may replace a rounded attention interval by an exact value using Theorem 3. Otherwise uncertainty may propagate through further sound operations, until each live write is known exactly or the attempt is rejected. This topological induction supplies the interval-containment premise of Theorem 7; it is not a formal refinement proof of the Python interpreter.

Before execution, the caller retains the immutable original state. Every proposed write is staged privately; a reference-equal scalar is appended only to the staged representation. No reader may observe staging before the final decision is certified. On rejection the staged object is discarded, and dense evaluation starts from the original state, not from a partially modified cache. On acceptance all persistent updates are published as one logical transaction. Correct model/sequence attribution and a complete state inventory are premises of this publication step.

For future-trace composition, let $\hat{F}(C, x)$ be the filtered transition. The numerical and write gates establish $\hat{F}(C, x) = F_R(C, x)$ on accepted invocations, and fallback establishes the same equation on rejected invocations. For a supplied sequence $x_0, \dots, x_{T-1}$, the base states agree. If states at step j agree, the common input and the one-step equation give equal tokens and next states. Induction proves all T tokens and final states equal. For greedy generation replace the system state by (C, x) and use $(C, x) \mapsto (y, (C', y))$; equal emitted tokens provide equal next inputs. The statement concerns every finite legal continuation and contains no accumulated numerical error or failure probability.

D. Outward Exponential Enclosures and E24 Rounding

Let $\Delta = 2^{-24}$ and define $Q_\Delta(x) = \Delta \text{RNE}_{\mathbb{Z}}(x/\Delta)$ on nonnegative real inputs. Suppose a sound exponential computation gives $0 \leq L \leq e^z \leq U$. The quantized weight satisfies

$$E_{24}(z) \in [Q_\Delta(L), Q_\Delta(U)] \subseteq [\max(0, L - \Delta/2), U + \Delta/2]. \quad (20)$$

Thus rounding the outward enclosure endpoints to the target lattice is at least as strong as adding a global half-quantum error, provided the endpoint quantization itself is exact.

Proof. Nearest-even rounding is nondecreasing. To see this directly, suppose $x \leq y$ but their rounded lattice values satisfy $a > b$. Nearest-point optimality implies $|x - a| \leq |x - b|$ and $|y - b| \leq |y - a|$, giving $x \geq (a + b)/2 \geq y$. Hence $x = y$, whose deterministic rounding cannot give different outputs, a contradiction. Applying this monotonicity to $L \leq e^z \leq U$ proves the first inclusion. Every nonnegative input is within $\Delta/2$ of a nonnegative lattice point, so $|Q_\Delta(x) - x| \leq \Delta/2$; the second inclusion follows. $\square$

The implementation uses exact power-of-two scaling, integer parity, and nearest-even tie handling for admitted finite binary64 endpoints, rejecting overflow or invalid inputs. The generic monotone endpoint implication is covered by the compiled `StateCut.monotone_weight_bounds`; the binary64 implementation and exponential enclosure are separately tested, and are not formally refined to that theorem. An exact negative-tail case uses $z \leq -25$: since $e > 2$, $e^z \leq e^{-25} < 2^{-25} = \Delta/2$, so $E_{24}(z) = 0$. Neither construction applies a common score shift.

E. A Backend Bridge Is an Additional Obligation

Proposition 8 (Inward-shifted bridge). *Let $D > 0$, $A = N/D$, $\varepsilon \geq 0$, and suppose a backend’s pre-materialization scalar B satisfies $|B - A| \leq \varepsilon$. If residual bounds establish*

$$R(t_- + \varepsilon) \geq 0, \quad R(t_+ - \varepsilon) \leq 0, \quad (21)$$

with strict signs for excluded cell edges, then $B \in J_y$ and its materialization in the same rounding format is y .

Proof. The residual signs give $A \geq t_- + \varepsilon$ and $A \leq t_+ - \varepsilon$. Since $B \geq A - \varepsilon$ and $B \leq A + \varepsilon$, $t_- \leq B \leq t_+$. A strict first or second inequality gives the corresponding strict endpoint condition. These are exactly the requirements for membership in J_y . $\square$

The result does not construct ε . In particular it cannot be instantiated by an empirical tolerance, a few matching output tensors, or a theorem about a different exponential. If the inward-shifted cell is empty the sufficient conditions cannot pass, which correctly reflects the insufficiency of that error bound for exact rounding. A state-level bridge must cover every numerical operation contributing to a live write, not only the final attention quotient.

F. Reproducibility and Artifact Map

The repository retains raw machine-readable measurements and their logs. The manuscript’s generated tables contain no manually entered measured numbers; their input paths and SHA-256 hashes appear in `paper/generated/input_manifest.json`. Current GH200 measurements are retained under `results/gh200/`; historical release snapshots do not substitute for a successful current execution.

Artifact	Responsibility
<code>src/statecut/arithmetic.py</code>	Exact rounding, elementary enclosures, greedy tie rule
<code>src/statecut/residual.py</code>	Chord and finite-count residual predicates
<code>src/statecut/forest.py</code>	Persistent summary construction and refinement tree
<code>src/statecut/tree_attention.py</code>	Attention certificates and exact raw-leaf fallback
<code>src/statecut/tree_model.py</code>	Exact cuts and staged persistent-write checks
<code>scripts/run_v2_experiments.py</code>	Paired logical-read and complete-state experiments
<code>scripts/capture_pretrained.py</code> , <code>scripts/audit_capture.py</code>	Backend-preserving capture and offline audit
<code>lean/StateCut/</code>	Theorem sources with a pinned Lean/Mathlib environment
<code>docs/LITERATURE_AUDIT.md</code>	Primary-source comparison and novelty scope
<code>paper/</code>	ICML preprint source, official style, bibliography, and PDF

From the repository root, install the test dependencies and run:

```
python -m pip install -e '[test]'
python -m pytest -q
python scripts/run_v2_experiments.py \
    --output results/gh200/algorithm/v2_experiments.json
bash scripts/check_lean.sh
make -C paper
```

The Lean check must return success after compiling the actual declarations; source scans for forbidden placeholders are supporting checks only. CUDA build and capture commands, compiler flags, model revision, dtype, attention implementation, prompt token IDs, and device metadata belong in the corresponding run record. Capture tests can skip if optional Torch or NumPy packages are missing; both pass and skip counts must be inspected.

The official ICML 2026 style is vendored without modification from the conference’s author-instruction archive, whose source and hash are in `paper/template_source.json`. The manuscript uses the style’s `preprint` option.

G. Pretrained CUDA Domain Accounting

Block size	Unsupported weights	Positive mass unproved	Valid positive mass
1	346	4	10,558
16	415	242	10,251
64	462	234	10,212
128	524	224	10,160

The columns partition the query heads within each configuration. Unsupported weight enclosures are rejected before the output predicate; a nonpositive lower denominator bound does not establish that the actual denominator is nonpositive. The two proposal methods share these counts. Accepted-head counts use the full population, including rejected domains, rather than silently excluding unsupported cases. CUDA admits a narrower positive-score domain than the rational oracle. The capture audit requires the exact query scale $1/8$, as used by these head-dimension-64 models; arbitrary rounded query scaling is outside this audit’s contract.

H. Compiled Formal Inventory

The build uses `leanprover/lean4:v4.19.0`, with Mathlib at `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The audit counts source theorem declarations separately from generated logical helpers. It records 14 compiler runtime extraction auxiliaries outside the proof inventory; these are not used to replace kernel checking. The reported counts and names come from the build audit JSON.

Claim family	Compiled declarations in namespace StateCut
Moment and residual bounds	<code>abs_chord_sum</code> , <code>moment_residual_sound</code> , <code>finite_abs_sum_bound</code> , <code>finite_residual_bound</code>
Finite-count strength and feasibility	<code>finite_envelope_le_chord</code> , <code>finite_envelope_attained</code> , <code>floor_remainder_contract</code>
Rounding-cell composition	<code>residual_to_cell</code> , <code>moment_to_cell</code> , <code>finite_to_cell</code>
Conditional numerical bridge	<code>backend_bridge_gate</code> , <code>backend_bridge_cell</code>
Disjoint frontier sums	<code>disjoint_split_sum</code> , <code>frontier_sum_bounds</code>
Persistent writes and traces	<code>singleton_writes</code> , <code>checked_write_frontier</code> , <code>write_frontier_future</code>

The finite-count theorem takes an integer endpoint count and a remainder satisfying the summary equation. The separate mathematical floor/remainder theorem supplies those conditions in the nonconstant, non-upper-endpoint case. Constant cases are proved separately. Neither result establishes that Python’s runtime executes the specified mathematics; the exact-rational tests check that implementation separately. The cell theorems take a correct rounding-cell contract as a premise and do not prove the full BF16 encoding/cell generator. The frontier results concern disjoint sums; they do not formally verify the Python persistent forest or its memory-access counters.

I. Interpretation of Negative Results

An exact certificate is useful only if its premises can be established cheaply and it accepts often enough for the intended workload. Failure of an interval gate does not show that the proposal is wrong; it shows that the current enclosure cannot prove it right. Full raw fallback with exact state parity is therefore a valid correctness outcome and a negative performance outcome. Similarly, a zero-error device test is finite conformance evidence, while a counterexample can falsify a universal implementation claim immediately. The present artifact preserves both types of result.